Supplementary Information for: High Helium Retentivity in Goethite under Hydrothermal Conditions Measured by Step-Heat Experiments in Silica Glass Ampoules

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1. Derivations for Bounding Method Equations

The uncertainty of the helium evolution of the encapsulated natural material during a heating step (ω) due to permeation lag correction can be estimated by calculating the maximum possible helium residing in the tube between steps, i.e. for the values for $n_{0,}$ and n_t: n_0^{UB} and n_t^{UB}. From the permeation lag equation (main text Eq. (1)), upper bound estimates for $n_{0,}$ and n_t bound the encapsulated material contribution ω within:

$$m - n_0^{UB} \leq \omega \leq m + n_t^{UB} \tag{1}$$

Upper bounds for n_t for a given high temperature experimental step can be calculated if step heats are long enough to assume steady-state permeation, assuming G and K are known or can be conservatively estimated. To proceed, one must make the generally conservative assumption that the production rate of helium within the ampoule due to degassing of the encapsulated natural sample ($\Omega = \frac{\omega}{t}$) is constant over time at a given temperature. In reality, Ω will decrease over time for helium loss governed by first order processes (Yakubovich et al., 2010; Wilson and Bisson, 1973) or diffusion (except for the case of initially inhomogeneous distributions, the memory of which is often erased early in any diffusion experiment: Shuster et al., 2004).

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Eq. (3), given below, shows that the amount of helium trapped in the tube (n_t) is proportional to the rate of helium outgassing from the natural encapsulated sample (Ω), so the approximation that this rate is constant will overestimate the amount of unanalyzed helium remaining in the ampoule after a high temperature experimental step. The rate of helium degassing over time will decrease for degassing controlled by second-order volume diffusion *except* for the minor case of the alpha-ejection effect at low He-release fractions (Farley et al., 1996; Ketcham et al., 2011) or for the more serious effects of radiation damage annealing (Goldsmith et al., 2020) and mineral decomposition (Farley et al., 2024). In any case, sequential isothermal steps immediately reveal when $\frac{\omega}{t}$ increases over time.

A new first order differential equation like Eq. (3) is formulated, now incorporating a constant source term (Ω) to conservatively approximate the helium influx into the ampoule from the degassing natural sample.

$$\frac{dn}{dt} = \Omega - \lambda n \tag{2}$$

Where $\Omega = \frac{\omega}{t}$ and $\lambda = KGRT$. Immediately, the steady-state quantity of helium atoms (n_{ss}) present in the ampoule can be found by solving Eq. (2) for the steady-state condition where $\frac{dn}{dt} = 0$.

$$n_{ss} = \frac{\Omega}{\lambda} = \frac{\omega}{\lambda t} \tag{3}$$

Additionally, the helium concentration within the ampoule during the approach to the steady-state concentration can be solved for by integrating Eq. (2) with respect to t. First, the integrating factor for Eq. (2) is identified: $I = e^{\lambda t}$, which is multiplied to both sides to yield Eq. (4).

$$e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n = e^{\lambda t}\Omega \tag{4}$$

Then both sides are integrated with respect to *t:*

$$\int \left(e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n \right) dt = \int e^{\lambda t}\Omega dt$$

Because $\frac{d}{dt}(ne^{\lambda t}) = e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n$ by the product rule Eq. (4) reduces to:

$$n_t e^{\lambda t} = \frac{\Omega}{\lambda}e^{\lambda t} + C \tag{5}$$

Where C is an unknown constant introduced by indefinite integration. This expression becomes more intuitive when both sides are divided by the integration factor and by substituting the steady-state concentration from Eq. (3).

$$n_t = n_{ss} + Ce^{-\lambda t} \tag{6}$$

The integration constant C can be solved by enforcing the condition that $n_t(t = 0) = n_0$, leading to:

$$n_0 = n_{ss} + C \tag{7}$$

Solving Eq. (7) for C and substituting into Eq. (6) yields Eq. (8). Now an inequality is assigned to the equation to reflect the initial assumption that the isothermal helium outgassing rate (Ω) is constant *or decreasing*.

$$n_t \leq \ n_{ss} + (n_0 - n_{ss})\, e^{-\lambda t} \tag{8}$$

For long timescales (i.e. when $e^{-\lambda t} \approx 0$), Eq. (8) reduces to $n_t \leq \ n_{ss}$. This is effectively true for steps long enough such that $t \geq \frac{5}{\lambda}$, for which $e^{-\lambda t} \leq 0.007$. It is advisable to only consider the Approximation with Uncertainty method for steps of sufficiently long duration such that $t \geq \frac{5}{\lambda}$ so that the estimation for n_t can be reduced to simply the steady-state value and not be dependent upon the estimation for n_0 or its uncertainties.

To estimate the upper bound on the permeation lag (n_t^{UB}), one must consider the circumstances that would produce the highest possible value for n_t. In addition, the constraint that $t \geq \frac{5}{\lambda}$ is enforced so that previous heating steps are not significant, simplifying the problem significantly. Because n_t scales with ω by Eq. (3), the upper bound estimate n_t^{UB} must assume the upper bound estimate for ω, which is given in Eq. (1) as $m + n_t^{UB}$. Substituting $\omega^{UB} = m + n_t^{UB}$ into Eq. (3) to express n_t^{UB} in terms of the measured value m:

$$n_t^{UB} = n_{ss}(\omega = m + n_t^{UB}) = \frac{m + n_t^{UB}}{\lambda t}$$

Solving for n_t^{UB} yields Eq. (9):

$$n_t^{UB} = \frac{m}{\lambda t - 1} \tag{9}$$

2. Analytical Methods for Hydrothermal Goethite Step Heat Experiment

2.1. Uncertainty Quantification for Bounding Method

Helium-in-Goethite Diffusivity Inversions. Apparent values of D/a^2 for ^{3}He and ^{4}He were calculated from times and cumulative measured yields using the equations of Fechtig and Kalbitzer (1966). Lower and upper bounds for diffusivity were calculated using the upper bound for the helium contents of the ampoules before and after each step, evaluated using (8). Diffusivity for step i is a function of the cumulative fractional yield for step i $\Sigma^i F$ and the cumulative fractional yield of the previous step $\Sigma^{i-1} F$. The diffusivity is maximized when the difference between the fractional yields is largest, and minimized when the difference is smallest. This difference is maximized for the extreme case where the previous step ended with no permeation lag, and the current step ended with the maximum amount of permeation lag ($^i_{SS}n_t$), whereas the opposite is true for the minimum diffusivity.

$$\begin{aligned} {}^{UB}D_\omega &= D\Big(\Sigma^{i-1}F_m,\ \Sigma^i F_m + {}^i_{SS}n_t\Big) \\ {}^{LB}D_\omega &= D\Big(\Sigma^{i-1}F_m + {}^{i-1}_{SS}n_t,\ \Sigma^i F_m\Big) \end{aligned} \tag{10}$$

For the ^{3}He diffusivities, a linear regression was fit to the steps with cumulative ^{3}He fractional yield $\geq$7% (12 steps, Fig. 7), yielding an R^2 statistic of 0.989. The fit's slope and intercept are used as a reference line to calculate "Delta" values for the diffusivities of ^{3}He and ^{4}He (Fig. 9). No uncertainty on the total amounts of ^{3}He and ^{4}He are propagated into the diffusivities because the experiment operated at low cumulative yields where this error affects all diffusivities approximately homogeneously, so it is only relevant for the y-intercept of the regression and the corresponding calculated closure temperature. The uncertainty from the intercept and slope parameters of the linear regression is significantly larger than the uncertainty related to the total yield, so it is neglected.

Helium Isotope Ratio Uncertainty. The upper and lower bounds of the $\frac{^4He}{^3He}$ isotope ratios for each heating step are taken as the maximum and minimum $\frac{^4He}{^3He}$ isotope ratio calculated out of the 4 hypothetical end-member permeation scenarios, which would have been experienced by both helium isotope systems:

1. $n_t = 0,\ n_0 = 0$

2. $n_t = n_t^{\mathrm{UB}},\ n_0 = 0$
3. $n_t = n_t^{\mathrm{UB}},\ n_0 = n_0^{\mathrm{UB}}$
4. $n_t = 0,\ n_0 = n_0^{\mathrm{UB}}$

Intuitively, higher differential permeability between the two helium isotopes ($\delta = \frac{K_{3He}}{K_{4He}}$ here) will result in greater isotope fractionation between the measured $\left(\frac{m_4}{m_3}\right)$ and true $\left(\frac{\omega_4}{\omega_3}\right)$. Furthermore, lower absolute permeabilities will result in stronger permeation lag, additionally producing higher isotope fractionation in the measured value. The ^{4}He permeability is thus conservatively taken to be identical to Swets et al. (1961), an anhydrous literature value which this experiment's own isothermal ^{3}He permeation measurements show to underestimate the ampoule's true permeability (Fig. 5); ^{4}He permeability itself could not be independently measured in this experiment. For the purposes of evaluating the measured isotope ratios, the relative ^{3}He/^{4}He permeability (δ) is taken to be 1.29, the 2σ upper bound of the value measured independently for a helium-soaked water-filled ampoule: 1.10 ± 0.19 (2σ). 1.29 is in great excess of realistic values of the relative ^{3}He/^{4}He permeability, which is expected to be between 1.06–1.11 for the temperature range relevant to this experiment (Jones, 1953). These conservative assumptions taken together guarantee that the calculated upper and lower bounds for the isotope ratios accurately constrain the true value. The upper bounds of the ampoule contents are calculated using the steady-state approximation of Eq. (8) (main text §2.6.5): $n_{ss} = \frac{m}{\lambda t - 1}$. For the four extreme cases listed above, the following equations apply:

1. $\dfrac{\omega_4}{\omega_3} = \dfrac{m_4}{m_3}$
2. $\dfrac{\omega_4}{\omega_3} = \dfrac{m_4 + \frac{m_4}{\lambda t - 1}}{m_3 + \frac{m_3}{\delta \lambda t - 1}}$
3. $\dfrac{\omega_4}{\omega_3} = \dfrac{m_4 + \frac{m_4}{\lambda t - 1} - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 + \frac{m_3}{\delta \lambda t - 1} - \frac{m_3^{i-1}}{\delta (\lambda t)^{i-1} - 1}}$
4. $\dfrac{\omega_4}{\omega_3} = \dfrac{m_4 - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 - \frac{m_3^{i-1}}{\delta (\lambda t)^{i-1} - 1}}$

Where m_x^{i-1} and $(\lambda t)^{i-1}$ refer to the measured helium content and λt of the previous step, respectively. λ here applies explicitly to ^{4}He, whereas $\delta\lambda$ applies to ^{3}He. The upper and lower bounds of the isotope ratios are then taken to be the maximum and minimum of these four extreme cases.

2.2. Sensitivity and Uncertainty Calculations

^{3}He and ^{4}He sensitivities were calculated for each analysis of the MM standard gas, assuming that each standard aliquot contained 157.50 ± 0.41 (2σ) nccSTP ^{4}He and 3.567 ± 0.040 (2σ) pccSTP ^{3}He. These values were obtained from sensitivities from a known amount of a standard gas with similar helium abundances and from the signals from 10 aliquots of the new MM standard tank in June 2023.

1σ uncertainties related to internal reproducibility were estimated for each sensitivity calculation as the rolling standard deviation of sensitivities calculated from 5 MM standards. In other words, the internal reproducibility of the sensitivity was estimated as the standard deviation of each calculated sensitivity and the 2 sensitivities immediately before and after. Uncertainties related to external reproducibility from the amount of ^{3}He and ^{4}He in each standard aliquot were added to the total uncertainty as well.

Sensitivities and their uncertainties were interpolated to times between standard analyses using a Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) from the SciPy package in python. These interpolated sensitivities and uncertainties were used to convert the ^{3}He and ^{4}He signals (cps and fA respectively) to absolute atomic abundances and uncertainties in units of pccSTP and nccSTP respectively. The sensitivities and interpolations are graphically displayed in the "STDs" tab in a plot generated using the Matplotlib package in python.

3. Finite Difference Diffusion Model of Cooling Tube

3.1. Numerical Methods

A numerical model to estimate diffusive permeation is invaluable, as it provides a flexible way to estimate the permeation fraction $\left(\frac{m}{n_0}\right)$ even when the required assumptions for the analytical solutions are not met. For example, a numerical approach can calculate the expected permeation fraction for a non-isothermal time-temperature history, and it needn't obey the assumption that the helium concentration profile in the glass is in diffusive equilibrium with the partial pressure of helium in the encapsulated gas phase. As

will be investigated below, the equilibrium concentration profile assumption is ephemerally broken when temperatures are changed, and for ampoules with high glass/gas ratios it is never true.

Carslaw and Jaeger (1959) give the equation for the change in concentration by diffusion in a cylinder (eqn. 5, page 17, second edition), expressed here in terms of concentration and diffusivity.

$$\frac{dC}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r} + \frac{1}{r^2}\frac{\partial^2 C}{\partial \theta^2} + \frac{\partial^2 C}{\partial z^2}\right] \tag{11}$$

For typical ampoules used in this work, the surface area of the top and bottom together is about 5–10% of the surface area of the walls, but they are neglected here for simplicity. Assuming that the concentration profiles are radially symmetrical and uniform with height (z), Eq. (11) is reduced to the form used throughout this section:

$$\frac{dC}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r}\right] \tag{12}$$

We numerically solve Eq. (12) using the Forward-Time Central-Space (FTCS) Finite Difference Method (FDM) along a radial transect of a model ampoule of arbitrary geometry for small, discrete time intervals. The system is spatially discretized into 100 cells along a radial transect, and numerical stability was ensured by solving for the time interval of a single step, Δt, such that $\frac{D\Delta t}{\Delta r^2} = 0.5$, satisfying the Courant-Friedrichs-Lewy (CFL) condition.

Helium concentrations along the transect (except for the boundaries, where r = a and r = b) are updated at each step using an FTCS discretization of the radially symmetric diffusion equation for a cylinder (Eq. (12)), repeated below for reference before its discretized form:

$$\frac{dC_r}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r}\right]$$

$$\Delta C_r \approx \Delta t \bullet D\left[\frac{C_{r-\Delta r} - 2C_r + C_{r+\Delta r}}{\Delta r^2} + \frac{1}{r}\frac{C_{r+\Delta r} - C_{r-\Delta r}}{2\Delta r}\right] \tag{13}$$

Dirichlet boundary conditions were enforced at the inner and outer boundaries. The concentration at the model ampoule's outer boundary (where r = b) was held at 0, while the concentration at the inner boundary (where r = a) was calculated at each discrete step by multiplying the solubility (in units

of $\frac{atoms}{atm \times cm^3}$) by the helium partial pressure within the ampoule, according to Eq. (14) or (15).

$$C_{He}^{glass} = S(T) \bullet P_{He}^{gas} \tag{14}$$

Eq. (14) can be related to the atomic abundance of helium in the gas phase using the ideal gas law and the formula for the volume of a cylinder:

$$P_{He}^{gas} = \frac{n_{gas} RT}{V_{gas}}$$

$$V_{gas} = \pi a^2 h$$

$$C_{He}^{glass} = S(T) \bullet \frac{n_{gas} RT}{\pi a^2 h} \tag{15}$$

For helium permeation out of glass ampoules, the internal pressure is not constant. n_{gas} changes after every step and Eq. (14) is reapplied to the inner boundary cell where r = a. While the helium concentrations in the glass and in the encapsulated gas phase are both dependent on the length of the ampoule (h), they compensate for one another such that the absolute atom flux into and out of the ampoule is not a function of h.

In order to simulate permeation, the atomic abundances of helium within the gas phases both outside the ampoule (at high vacuum) and inside the ampoule must be tracked, as do the fluxes between the glass and these two gas phases. These fluxes were calculated simply as the diffusive flux in $\frac{atoms}{s \bullet cm^2}$ as is given by Fick's law. For spatially discrete cells throughout the ampoule at any radius except for the boundaries, the diffusive flux can be approximated with the finite difference:

$$J_r = -D \frac{\partial C_r}{\partial r} \approx -D \frac{C_{r+\Delta r} - C_{r-\Delta r}}{2\Delta r} \tag{16}$$

However, Eq. (16) cannot be applied at the ampoule's inner wall when r = a or outer wall when r = b because there is no remaining glass on one side of the cell. In that case, second order terms from Taylor expansions centered at $a + \Delta r$ and $b - \Delta r$ can be used to better approximate the $\frac{\partial C_a}{\partial r}$ and $\frac{\partial C_b}{\partial r}$ terms. Letting $f(r) = \frac{\partial C_r}{\partial r}$ and $f'(r) = \frac{\partial^2 C_r}{\partial r^2}$:

$$J_a = -D\frac{\partial C_a}{\partial r} \approx -D\left(\frac{\partial C_{a+\Delta r}}{\partial r} + \frac{\partial^2 C_{a+\Delta r}}{\partial r^2}\Delta r\right) \approx -D\left(\frac{C_{a+2\Delta r} - C_a}{2\Delta r} + \frac{C_a - 2C_{a+\Delta r} + C_{a+2\Delta r}}{2\Delta r^2}\Delta r\right) \tag{17}$$

$$J_b = -D\frac{\partial C_b}{\partial r} \approx -D\left(\frac{\partial C_{b-\Delta r}}{\partial r} - \frac{\partial^2 C_{b-\Delta r}}{\partial r^2}\Delta r\right) \approx -D\left(\frac{C_b - C_{b-2\Delta r}}{2\Delta r} - \frac{C_b - 2C_{b-\Delta r} + C_{b-2\Delta r}}{2\Delta r^2}\Delta r\right) \tag{18}$$

The helium contents in the measurable (external, high vacuum) reservoir and the inner encapsulated gas phase are updated every step using fluxes determined from Eq. (18) and (17) respectively. To convert these area-normalized fluxes into absolute atomic abundances, these fluxes are multiplied by the time increment (Δt) and the surface area of the cylinder at the outer and inner walls respectively.

Simulations require an initial helium concentration profile in the glass appropriate for a fresh charge of helium at a given temperature. Rather than deriving a closed-form analytical solution for this profile, it is obtained numerically: the inner and outer boundary conditions are held fixed at their values for that charge and temperature (Eq. (15) and 0, respectively) while the model iterates Eq. (13) until the concentration profile no longer changes, i.e. a numerical 'burn-in' to steady state. This produces the same result as solving Eq. (12) analytically for the steady-state profile, without requiring a separate closed-form derivation.

3.2. A Parameterized Cooling Rate of the HT-Win-06-01b Ampoule

One potential concern about the results of the internally measured permeabilities is that some additional helium was permeated during the cooling period after each permeation step. In order to evaluate the significance of this effect, a finite difference model was built in python to simulate diffusion, influx, and outflux of helium across a hollow glass cylinder. The glass cylinder was given the same dimensions of the ampoule used in the Win-06-01b hydrothermal experiment, and was subjected to the same heating and cooling schedule as the hydrothermal experiment's permeation steps. The tube was charged with 100 arbitrary units of helium atoms at the end of each experimental step, and the concentration profile for the new arbitrary charge was initialized at T = 24°C using the numerical burn-in procedure described in §3.1. Permeability values for ^{3}He were assumed as the product of the ^{4}He

permeability values given by Swets et al. (1961) and the ^{3}He/^{4}He relative permeabilities given by Jones (1953).

To simulate cooling, a cooling rate had to be assumed. To most accurately simulate the cooling of the experimental ampoule, Watlow temperature records of the experimental ampoule during cooling steps were used to calculate the cooling rate dT/dt for discrete time intervals. A quadratic function was fit to these data points to model the cooling rate as a function of temperature with a single root at T = 24°C such that the temperature would stabilize at T = 24°C by design.

$$\frac{dT}{dt} = -\left(9.23081 \times 10^{-6}\right)(T-24)^2 - \left(7.74304 \times 10^{-4}\right)(T-24)$$

(dT/dt in °C/s, T in °C; $R^2 = 0.9633$).

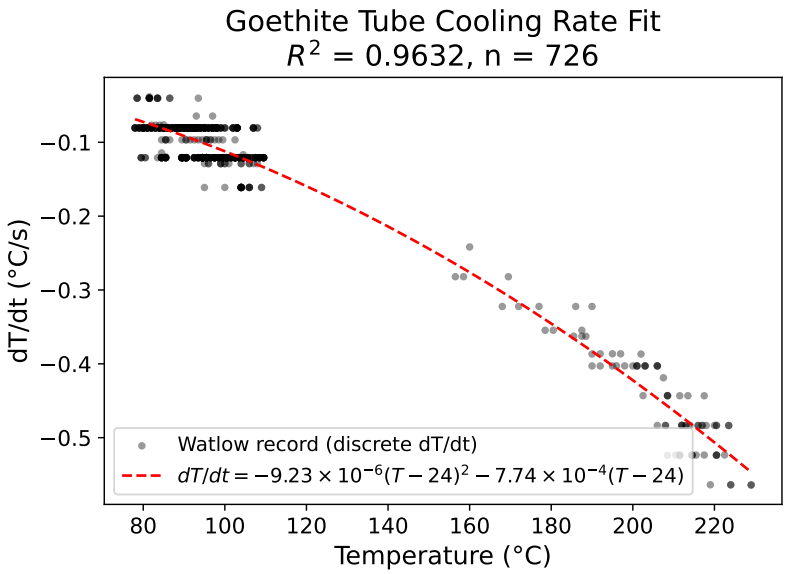


Figure S1: Discrete cooling rate (dT/dt) values calculated from the Watlow temperature record for the Winsor Goethite experiment, plotted against temperature, with the quadratic fit used to simulate cooling in the finite difference diffusion model overlain.

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